

Request for White Paper (RWP) High-Performance Compressor System Prototype

Issuing Office: Air Force Test Center, Installation Contracting Division (AFTC/PZI)

Response Due Date: No later than 1:00 PM Central Time, 8 weeks after the date of posting.

1.0 PURPOSE

The Air Force Test Center's Installation Contracting Division (AFTC/PZI) at Arnold Air Force Base (AFB) is seeking White Papers from interested parties capable of meeting the Government's requirement for an Analysis of Alternatives and the design, manufacture, and testing of a Prototype High-Performance Compressor System. This effort is issued under the authority of 10 U.S.C. § 4022 for Other Transaction Prototype (OTP) agreements.

The primary goal of this RWP is to identify potential solutions and innovative approaches to meet the technical objectives detailed in the attached Statement of Objectives (SOO), Attachment 1. The focus of the white paper should be on a comprehensive approach that validates critical design elements and ensures the successful integration and long-term sustainment of the final system. The Government will use the submitted White Papers to assess industry capabilities, refine project requirements, and select vendors for a potential future Request for Prototype Proposal (RPP).

2.0 BACKGROUND

The Arnold Engineering Development Complex (AEDC) provides the United States with a critical capability for the testing and evaluation of aerospace systems. The successful development of future capabilities relies on ground-test facilities that can accurately replicate extreme flight environments. The current project addresses a capability gap for a high-pressure, high-mass-flow air source. The objective is to develop a system that meets the performance requirements detailed in the SOO.

3.0 PROJECT OBJECTIVES & DEFINITION OF SUCCESS

A successful prototype effort will be one that comprehensively de-risks the full project by validating the operational utility of the contractors proposed design. Operational utility, within the Department of War, refers to the degree to which a system, weapon, technology, capability, or process contributes to the successful completion of military missions and objectives within the operational environment. It encompasses mission effectiveness, integration, logistics, cost effectiveness, and adaptability.

The project is structured in three phases. The Government is seeking responses that address the full scope, though the execution of optional phases will be determined later.

- Phase I (Base Effort): System Analysis and Design
 - Conduct a comprehensive analysis and produce a complete system design, including a Process and Instrumentation Diagram (P&ID), a digital model, and an Analysis of Alternatives (AoA).

- Phase II (Optional Effort 1): Prototype Manufacturing and Test
 - Design, build, and factory-test a limited number of unique prototype compressors identified in the Phase I design to validate manufacturability and performance.
- Phase III (Optional Effort 2): Installation and Test Support
 - Provide on-site technical support to AEDC for the installation, commissioning, and operational testing of the prototype compressor(s).

Submissions should address how the proposed solution will meet the following criteria, which define a successful prototype:

1. Mission Effectiveness:

The prototype must validate the core performance of the design.

- **Performance Validation:** Through factory testing, the prototype must prove that the system design can achieve the performance points within the envelope (mass flow and pressure) as specified in SOO Section 3.1.
- **Operational Requirements:** The prototype's performance data must confirm the system's ability to meet all operational requirements as detailed in SOO Section 3.2.

2. Integration:

The prototype effort must produce a clear and viable path for integrating the full system.

- **Comprehensive Design:** The project must yield a complete system design package, including a final Process and Instrumentation Diagram (P&ID) and a functional digital model, which demonstrates how the full compressor layout will meet all performance requirements (Phase I). The P&ID shall conform to ANSI/ISA-5.1 standards or an equivalent, Government-approved standard.
- **Technical Support:** The contractor must provide effective on-site engineering support during the government-led installation and testing phase, ensuring a smooth transfer of technology and knowledge (Phase III).

3. Logistics & Supportability:

The prototype phase must generate the data and documentation necessary for long-term sustainment.

- **Technical Data Package:** A complete Level 3 Technical Data Package (TDP) must be delivered, containing all engineering drawings, design calculations, maintenance manuals, and a recommended spare parts list (Phase II).
- **Life-Cycle Analysis:** The project must deliver a credible analysis of estimated maintenance costs over a 20-year life, based on the operational usage specified in the SOO (Phase I).

4. Cost-Effectiveness:

The prototype must validate the financial viability and efficiency of the proposed solution.

- **Validated Costing:** The Analysis of Alternatives (AoA) must provide a data-driven comparison of drivetrain options based on life-cycle cost, and the prototype

manufacturing effort must serve to validate and refine the cost estimates for the full system (Phase I & II).

- **Government Purpose Rights:** The Government must receive Government Purpose Rights to all technical data, drawings, and models, ensuring future competitive procurement and sustainment without restriction (SOO Section 5.0).

5. Adaptability:

The project must honestly assess technological limits and provide viable contingency plans.

- **Envelope Feasibility:** The system analysis must determine if any part of the performance envelope is unachievable with current technology. If so, it must define the maximum possible envelope and propose options for technology maturation (Phase I).
- **Contingency Design:** If continuous runtime cannot be achieved, a fully designed High-Pressure Air (HPA) bottle farm alternative must be provided, complete with its own performance metrics (operation/recharge time) and risk analysis (SOO Section 4.1).

In summary, a successful prototype is not just a piece of hardware; it is the comprehensive validation of the system's performance, integration plan, cost, and long-term supportability.

Affordability: The Government estimates a total prototype project price between \$500k (Phase I) and \$50.7M (inclusive of Phases I, II, and III). This information is for planning purposes only and should not be interpreted as a cost ceiling or floor.

4.0 TECHNICAL OBJECTIVES

The core objective is to design a compressor system that meets the performance envelope specified in the SOO (Attachment 1).

5.0 WHITE PAPER SUBMISSION REQUIREMENTS

Respondents shall submit a White Paper limited to 20 pages. The response should include the following sections:

5.1. Company Information: Company Name, Address, Website, POC, CAGE Code, and SAM UEI.

5.2. Technical Approach & Prototype Definition:

- Describe your proposed design for the compressor system.
- Define your proposed prototype. Explain how a production or limited-run unit will be used to validate the high-risk components of the full design.
- Detail the plan for factory testing to validate key performance parameters.

5.3. Integration, Logistics, and Data Rights

- Outline your methodology for developing the final system design package and digital model.
- Describe your approach to providing on-site engineering support (Phase III).

- Detail your process for generating the Level 3 TDP and the 20-year life-cycle cost analysis. Detail any concerns with developing a Level 3 TDP, recommend other options if required.
- Confirm your understanding and compliance with the requirement to provide Government Purpose Rights to the Government. Discuss where this may be an issue and options to move forward.

5.4. Risk & Adaptability

- Provide your initial assessment of the most significant technical risks.
- Discuss your approach to determining the feasibility of the performance envelope and your strategy for developing maturation options if needed.
- If required, present a preliminary concept for the contingent HPA bottle farm design.

5.5. Corporate Experience

- Provide a summary of your company's experience in designing, manufacturing, and sustaining complex, high-performance rotating machinery and industrial systems.

5.6. Schedule: A notional schedule with major milestones for all three project phases.

5.7. Cost Estimate: A Rough Order of Magnitude (ROM) cost estimate for each project phase. If possible provide in Work Breakdown Structure (WBS) Level 2 or higher format.

5.8. Questions and Feedback: Provide assumptions that were used in the response. Provide requests for any missing information needed to review, design, and deliver the system. Provide any questions, comments, or feedback.

5.9. Compliance with 10 U.S.C. § 4022: Provide rationale and supporting documentation justifying how the company will satisfy at least one of the conditions for award under this authority. To satisfy 10 U.S.C. 4022(d)(1)(A)-(C) – Appropriate Use of Authority, elaborate on **one** of the following:

(A) There is at least one nontraditional defense contractor or nonprofit research institution participating to a significant extent in the prototype project.

(B) All significant participants in the transaction other than the Federal Government are small businesses (including small businesses participating in a program described under section 9 of the Small Business Act (15 U.S.C. 638)) or nontraditional defense contractors.

(C) At least one third of the total cost of the prototype project is to be paid out of funds provided by sources other than the Federal Government.

NOTE 1: Non-Traditional Defense Contractor (NDC) = An entity that is not currently performing and has not performed, for at least the one-year period preceding the solicitation of sources by DoD for the procurement or transaction, any contract or subcontract for the DoD that is subject to full coverage under the cost accounting standards (CAS) prescribed pursuant to

section 1502 of title 41 and the regulations implementing such section (see 10 U.S.C. §3014). A nontraditional defense contractor is considered a nontraditional defense contractor for the duration of the prototype project/agreement's period of performance, or for any in-scope work modified under the same agreement.

NOTE 2: Non-profit research institution is validated by reviewing the vendor's 501(c)(3) tax-exempt status, which is required by the Internal Revenue Service.

NOTE 3: "Significant extent" and "significant participants" may be defined by the vendor providing rationale to one or more of the following variables:

- a. Has a significant role / level of involvement in the development, production, or delivery of a product or service.
- b. Has a significant impact on the design, development, or production of the product or service.
- c. Has significant level of material interest in the success of the product or service.
- d. Has significant level of company's financial interest in the success of the product or service.
- e. Has Significant level of control over the development, production, or delivery of the product or service.
- f. Other significant rationale provided by the vendor.

6.0 EVALUATION CRITERIA

White papers will be evaluated based on the clarity of the proposed approach, the understanding of the project objectives, and the credibility of the path forward to delivering a system with high operational utility. The Government may invite select respondents to present their concepts in more detail.

7.0 ADMINISTRATIVE INFORMATION

Submissions: Responses shall be submitted in .pdf format via email to the Agreements Officer, Brooke Vandeman, at brooke.vandeman@us.af.mil.

Proprietary information should be minimized and **MUST BE CLEARLY MARKED**.

CONTRACTING OFFICE ADDRESS:

AFTC/PZI – "Notice ID# & Notice Name"
Attention: Brooke Vandeman
100 Kindel Drive, Suite A-332
Arnold AFB TN 37389-1335

Attachments:

Attachment 1: Statement of Objectives_HPC_Distro C EXPORT CONTROLLED

Note on Attachment 1: The SOO is Controlled Unclassified Information (CUI), Distribution C, and Export Controlled. To receive a copy, vendors must have a current (no more than three years old from the posting of this request) basic NIST SP 800-171 assessment in the Supplier Performance Risk System (SPRS) and be a certified contractor in the Joint Certification Program (JCP). Please contact the Agreements Officer to request the document and verify certification.